

RÓMULO GIANCARLO COLORADO BALBOA

Mechatronics Engineer | Manufacturing Data | Automation | Industrial Vision | Production Engineering

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PROFESSIONAL SUMMARY

Entry-level Mechatronics Engineer with experience in industrial vision, Python-based engineering tools, automotive manufacturing support, production tracking, technical documentation, and cross-functional coordination. Skilled in applying Python, OpenCV, manufacturing KPIs, automation logic, and process improvement concepts to support production, quality, validation, and engineering decision-making.

TECHNICAL SKILLS

Programming & Data: Python, C++, Linux command line, MATLAB, Simulink, pandas, Excel, data analysis, log analysis, technical reports

Manufacturing & Quality: OEE, scrap analysis, downtime analysis, production tracking, Six Sigma, process improvement, validation support

Computer Vision: OpenCV, image preprocessing, segmentation, edge detection, contour analysis, camera and lighting setup, automated inspection

Automation & Industrial Systems: PLC Siemens, TIA Portal, Ladder logic, sensors, actuators, SAP, Teamcenter, MES, FANUC, RobotStudio

Engineering Tools: SolidWorks, Fusion 360, CATIA, AutoCAD, ANSYS, Simscape

PROFESSIONAL EXPERIENCE

RECYCTUM

January 2025 – December 2025

Industrial Vision Developer

Puebla, Mexico

- Improved automatic material inspection and classification, achieving approximately 95 % accuracy through industrial vision systems developed with Python and OpenCV.
- Reduced material classification time by approximately 40 % using image processing algorithms, segmentation, edge detection, and contour analysis.
- Integrated cameras, lighting, and acquisition hardware to validate and optimize artificial vision solutions for industrial applications.

FORVIA

January 2024 – December 2024

Program Management Assistant – Lucid Air Serial Life Program

Puebla, Mexico

- Supported automotive program execution by managing schedules, engineering releases, prototype manufacturing, validations, and delivery milestones.
- Coordinated 40 to 60 part numbers and 80 to 100 engineering changes with internal teams, suppliers, and cross-functional stakeholders.
- Reduced prototype delivery time by 15 % through control dashboards, daily follow-up, priority tracking, and timely escalation of risks.

ENGINEERING PORTFOLIO PROJECTS

Manufacturing OEE Dashboard

Python, Streamlit, pandas, Plotly, Manufacturing KPIs

github.com/rgcb01/manufacturing-oe-dash-board

- Built a manufacturing analytics dashboard using synthetic production data to analyze OEE, scrap rate, downtime, throughput, and shift performance.
- Created KPI calculations, Pareto charts, filtered production views, and visual reports to support production monitoring and continuous improvement.

Industrial Vision and Embedded Prototype

Python, OpenCV, Raspberry Pi, MATLAB

- Developed a prototype for image acquisition, preprocessing, and inspection logic using Python, OpenCV, Raspberry Pi, and MATLAB.
- Applied image processing techniques to support automated classification and visual validation in an industrial-style workflow.

PLC Conveyor Automation Cell

PLC Siemens, TIA Portal, Ladder Logic

- Programmed a conveyor automation cell concept with sensor-actuator sequencing, operating states, basic fault response, and emergency-stop logic.
- Documented I/O logic, process sequence, and validation approach for industrial automation practice.

EDUCATION

Universidad Anáhuac Puebla

2019 – 2025

Bachelor's Degree in Mechatronics Engineering

Puebla, Mexico

Premio CENEVAL al Desempeño de Excelencia – EGEL, 2025

CERTIFICATIONS & DIGITAL CREDENTIALS

EF SET English Certificate – C2 Proficient, 87/100 | Six Sigma White Belt – CSSC | Modern Robotics – Northwestern University |

Autodesk Fusion 360 Manufacturing Process | MATLAB/Simulink/Simscape – MathWorks | Credly Digital Badges

LANGUAGES

Spanish: Native | **English:** C2 Proficient, EF SET 87/100 | **French:** A2 Basic